AGRIVIVE / MATKA-DRIP

HARYANA / PITCHER-IRRIGATION INSPIRED / CERAMIC SOIL TIP

SERIES 08 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



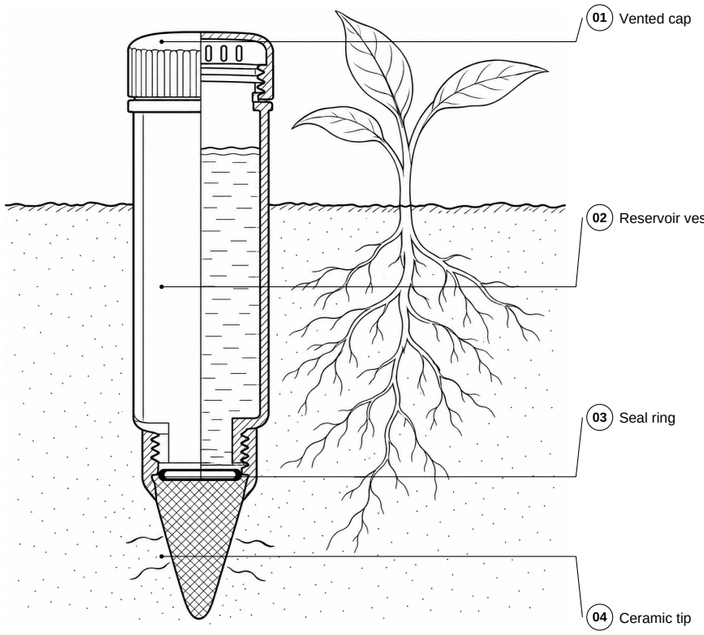
STATE CONTEXT / INDIGENOUS INSPIRATION

Buried porous clay vessels are used for irrigation in several parts of India. MATKA-DRIP adapts that shared practice to a refillable vessel with a replaceable ceramic tip; the practice is not exclusive to Haryana.

Practice source: CSIR: Buried clay pot irrigation for efficient and controlled water delivery | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 60 vessel OD x 200 overall H

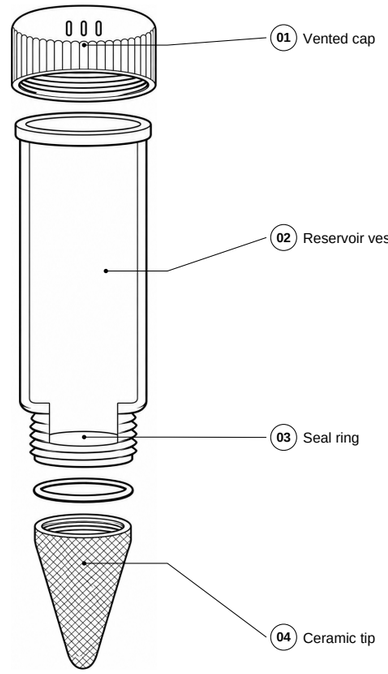


DESIGN / OPERATION NOTES

Seat ceramic tip near the trial root zone; protect it from impact.
Remove cap before filling, then refit the vented cap.
Water delivery depends on ceramic properties, soil suction and head.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
60 vessel OD x 200 overall H

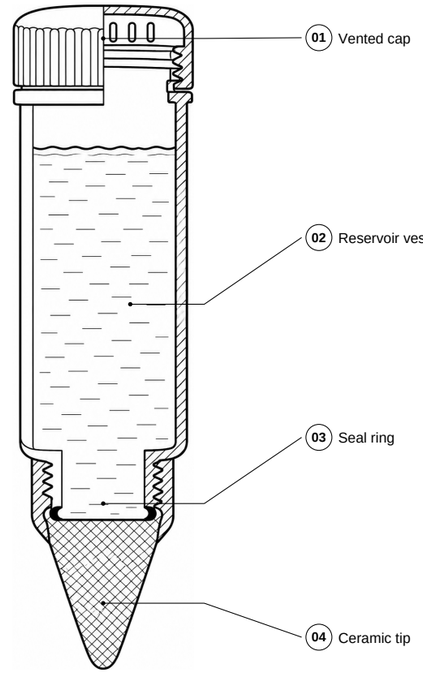


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

SECTION BASIS / mm
50 cavity ID x 140 clear H; 40 ceramic-tip length



DESIGN / OPERATION NOTES

Remove cap before filling, then refit the vented cap.
Water delivery depends on ceramic properties, soil suction and head.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

Vessel OD / overall H	**60 / 200**
Cavity ID / clear H	**50 / 140**
Ceramic-tip length	**40**

02 GEOMETRIC CHECK

GROSS WATER CAVITY

274.9 mL

pi/4 x 50^2 x 140 / 1000
Subtract internal fittings; seepage rate requires tests

03 PARTS / QTY

01 Vented cap	**1**
02 Reservoir vessel	**1**
03 Seal ring	**1**
04 Ceramic tip	**1**

Vessel/cap: candidate polymer. Tip: porous ceramic, pore grade TBD. Seal: elastomer grade TBD.

04 INTERFACES / CHECKS

Test ceramic permeability and mechanical strength.

Check seals, vent opening and refill access.

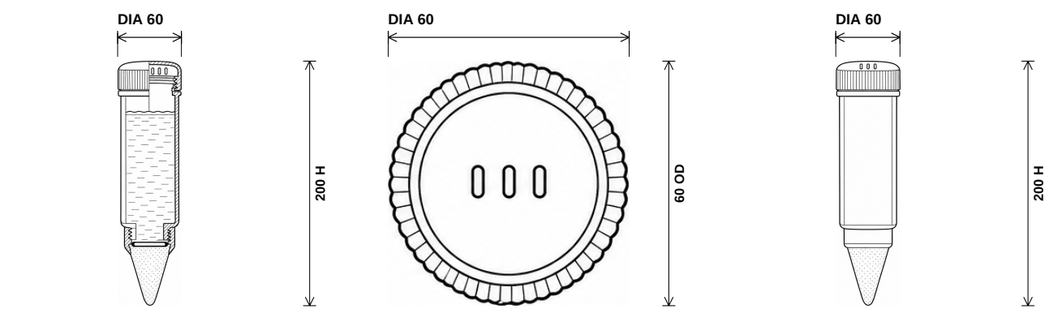
Measure usable volume and seepage under site conditions.

05 DATUMS / DRAWING BASIS

A: vessel axis. B: ceramic-tip seat. Keep refill neck and vent open during service.

PRELIMINARY / VALIDATION PENDING

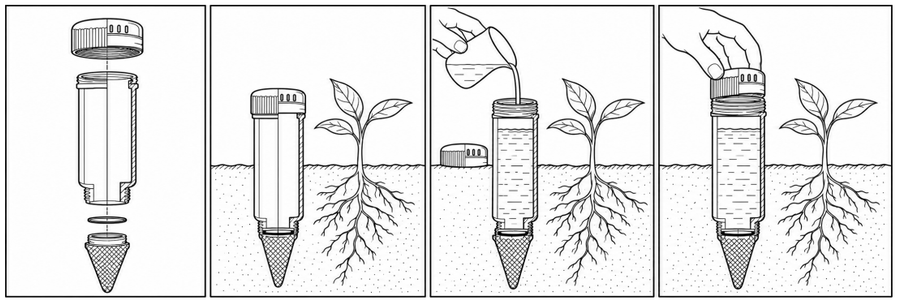
03 ORTHOGRAPHIC STUDIES / mm



FRONT SECTION **TOP** **SIDE**

Overall DIA 60 x 200 H. Cavity DIA 50 x 140 clear H; ceramic tip 40 L. Fits / cap detail TBD.

05 INSTALLATION / OPERATING SEQUENCE



1 FIT TIP / SEAL **2 SEAT BESIDE ROOT** **3 OPEN / REFILL** **4 REFIT CAP**

Concept sequence; validate fit, access and operating conditions in prototypes.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-08 / P01	MATKA-DRIP	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	08 / 28